



Punnett Squares Walkthrough Worksheet

Answers

1. In pea plants, purple flowers (F) are dominant and white flowers (f) are recessive. Parent one is FF and parent two is Ff.

	F	f
F	FF	Ff
F	FF	Ff

Key:
F- **Purple Flower**
f- **White Flower**

- a. What are the genotypes of the parents? Hint: Are they homozygous dominant, homozygous recessive or heterozygous?

Parent one - homozygous dominant, Parent two - heterozygous

- b. What are the phenotypes of the parents?

Both parents have purple flowers.

- c. What is the probability of each genotype of the offspring?

50% or $\frac{1}{2}$ homozygous dominant (FF), 50% or $\frac{1}{2}$ heterozygous (Ff)

- d. What is the probability of each phenotype of the offspring?

100% of the offspring will have purple flowers.

2. In pea plants, yellow seeds are dominant (T) and green seeds are recessive (t). Parent one is Tt and parent two is tt.

	t	t
T	Tt	Tt
t	tt	tt

Key:
T- **Yellow seed**
t- **Green seed**

- a. What are the genotypes of the parents?

Parent one - heterozygous, Parent two - homozygous recessive

- b. What are the phenotypes of the parents?

Parent one - yellow seed, Parent two - green seed

- c. What is the probability of each genotype of the offspring?

50% or $\frac{1}{2}$ heterozygous (Tt), 50% or $\frac{1}{2}$ homozygous recessive (tt)



- d. What is the probability of each phenotype of the offspring?

50% or $\frac{1}{2}$ yellow, 50% or $\frac{1}{2}$ green

3. In pea plants, round seeds are dominant (R) and wrinkled seeds are recessive (r). Parent one is RR and parent two is rr.

	r	r	
R	Rr	Rr	Key: T- Round seed t- Wrinkled seed
R	Rr	Rr	

- a. What are the genotypes of the parents?

Parent one - homozygous dominant, Parent two - homozygous recessive

- b. What are the phenotypes of the parents?

Parent one - round seed, Parent two - wrinkled seed

- c. What is the probability of each genotype of the offspring?

100% heterozygous

- d. What is the probability of each phenotype of the offspring?

100% round